ACADEMIC CAPSTONE PROJECT PRESENTATION

SMARTBIN

Smart Liquidation Bin Store E-Commerce Management System

A full-stack web platform for reverse logistics, dynamic daily price drops, physical bin tracking, and multi-tier retail operations.

- Project Domain: Reverse Logistics & Discount E-Commerce Retail
- **Technology Stack: Node.js, Express.js, MySQL 8.0, HTML5, CSS3, JavaScript, Bootstrap 5, Razorpay**
- **Target Users: Customers, Store Managers, and Super Administrators**

Project Overview & Background

- **The Liquidation Bin Retail Concept:** Liquidation stores purchase bulk pallets of overstock, customer returns, and clearance merchandise from major retailers (Amazon, Walmart, Target) and resell them with steep discounts.
- **Descending Daily Price Drops:** Items follow a systematic weekly price drop cycle where prices decrease each consecutive day (e.g., Friday $14, Saturday $10, Sunday $8, Wednesday $2, Thursday $1) to clear stock rapidly.
- **The Reverse Logistics Bottleneck:** Billions of dollars in returned products sit idle in distribution warehouses due to the lack of an organized, digital retail cataloging mechanism.
- **Physical Store Challenges:** Merchandise is traditionally dumped into wooden bins where shoppers search blindly without knowing inventory counts, conditions, or bin locations.
- **The SmartBin Digital Solution:** A centralized web application that indexes products by physical bin, displays verified condition grades, automates daily price drops, and enables 24/7 online shopping.

Project Objectives & Scope

- **Automate Daily Price Calculations:** Develop a dynamic database schedule that recalculates item prices every midnight automatically, eliminating manual price re-labeling and pricing discrepancies.
- **Provide Complete Inventory Transparency:** Provide customers with real-time visibility into physical store bin coordinates (Bin A, Bin B, Bin C) and verified condition ratings (Brand New, Open Box, Refurbished, Clearance).
- **Implement Multi-Role Governance:** Architect isolated and authenticated workflows for Customers, Store Managers, and Super Administrators using JWT sessions and bcrypt password encryption.
- **Enable a Full E-Commerce Pipeline:** Deliver an end-to-end shopping journey including dynamic cart calculations, stock validation, promotional discount vouchers, and Razorpay payment gateway integration.
- **Gamify Customer Retention & Loyalty:** Incentivize repeat buyers via integrated promotional coupon codes and automated Lucky Draw raffle competitions with transparent public winner announcements.

Existing System vs. Proposed System

Limitations of Existing Traditional Bin Stores:

- Mandatory in-person visits with zero pre-visit visibility into available stock or incoming pallets.
- Labor-intensive manual price relabeling and shouting daily rates across crowded store aisles.
- Unsorted and disorganized wooden bins causing customer frustration and damaged merchandise.
- Severe cashier queues during peak hours, paper-based invoices, and higher billing error rates.

Innovations of Proposed SmartBin Platform:

- ✓ 24/7 online storefront allowing customers to browse, filter, reserve, and buy items anytime.
- ✓ Automated midnight dynamic pricing recalculations based on the weekly day-of-week schedule.
- ✓ Structured bin mapping (Bin A, Bin B) and verified condition grading (Brand New, Open Box).
- ✓ Secure digital checkout via Razorpay API, simulated card processing, and live order status tracking.

Core Features of SmartBin

- **Dynamic Daily Price Drop Engine:** Automatically evaluates current day of week against the daily price schedule to calculate today's deal price (e.g., Friday $14 ➔ Thursday $1) without staff intervention.
- **In-Store Bin Location & Condition Tagging:** Products are mapped to physical store bins (Bin A, Bin B, Bin C) and tagged with condition ratings (Brand New, Like New, Open Box, Refurbished, Clearance).
- **Multi-Tier Role Portals & Security:** Separated, authenticated portals for Customers, Store Managers, and Super Administrators protected by JWT session tokens and salted bcrypt password hashing.
- **Smart Cart, Wishlist & Stock Validation:** Real-time stock validation, persistent wishlist storage, instant quantity adjustments, and dynamic subtotal calculations prevent overselling.
- **Promotional Coupon Engine:** Admin-created discount vouchers with code validation, percentage discounts, minimum cart spend limits, and expiration date checks.
- **Automated Lucky Draw Raffles:** Gamified customer retention module where eligible buyers enter raffle draws and winners are announced automatically on a public leaderboard.

System Architecture & Execution Workflow

- **Phase 1: Ingestion & Inventory Setup:** Store Managers catalog incoming liquidation pallets, assign physical Bin Numbers (Bin A, B), set condition grades, and upload merchandise photos using Multer middleware.
- **Phase 2: Automated Dynamic Price Calculation:** Backend server evaluates the current calendar day against the MySQL daily price schedule. The active selling price is dynamically recalculated and applied across the catalog.
- **Phase 3: Customer Discovery & Cart Management:** Shoppers explore the storefront, filter by category, brand, condition, or bin number, save items to their wishlist, add units to cart, and apply discount vouchers.
- **Phase 4: Payment Gateway & Database Synchronization:** Transactions are completed via Razorpay or simulated card gateways. In an atomic transaction, the system creates order/payment records and decrements stock.
- **Phase 5: Fulfillment & Lucky Draw Participation:** Store Managers review the dispatch dashboard and update statuses (Pending ➔ Confirmed ➔ Shipped ➔ Delivered). Customer orders automatically enter weekly Lucky Draws.

Module Description & Role Responsibilities

- **Customer Portal (Storefront):**
  - User Registration & Login: JWT session token management and bcrypt encryption.
  - Catalog Browsing: Real-time filtering by Bin Number, Category, Brand, and Condition Grade.
  - Cart, Wishlist & Vouchers: Subtotal calculation, stock checks, and promo coupon discounts.
  - Order Tracking & Raffles: Self-service order status tracking and Lucky Draw winner boards.
- **Store Manager Portal:**
  - Product Management: Add, edit, restock, and deactivate cataloged items in real time.
  - Physical Bin Allotment: Assign and reallocate products to physical store bins (Bin A, B, C).
  - Condition Grading: Categorize items as Brand New, Open Box, Refurbished, or Clearance.
  - Order Fulfillment: Review customer delivery details and update order dispatch stages.
- **Super Administrator Dashboard:**
  - Executive Analytics: Monitor revenue, orders, active managers, and registered customers.
  - Manager Governance: Onboard, update credentials, and manage privileges of store managers.
  - Price Schedule Master: Configure weekly liquidation price drop rules (Monday to Sunday).
  - Marketing & Raffles: Launch coupon campaigns and trigger automated lucky draw winner selection.

Database Design & Schema Architecture

13 Normalized Relational Tables in MySQL (smartbin_db / liquidation_bin):

- **Authentication & Roles Entities:** admins, managers, and users tables store user profiles with strict email uniqueness and 60-character bcrypt salted password hashing.
- **Inventory & Dynamic Pricing Entities:** products table stores bin_number, condition_type (ENUM), price, quantity, and images; daily_price_schedule stores weekly price drop values from Sunday to Saturday.
- **Commerce & Transactional Entities:** cart, wishlist, orders, order_items, and payments tables maintain complete transactional integrity with cascading foreign keys to users and products.
- **Gamification & Promotional Entities:** promotional_offers table validates coupon codes, discount percentages, and expiry; lucky_draws and lucky_draw_winners tables record raffle contests and winner history.
- **Database Normalization & Integrity:** Structured in Third Normal Form (3NF) to eliminate data redundancy and ensure ACID transactional consistency.

Technology Stack & Development Tools

- **Frontend Layer:** HTML5 for accessible semantic layout, CSS3 for custom styling and responsive media queries, Bootstrap 5 for responsive grid systems and modals, Vanilla JavaScript (ES6+) for dynamic DOM rendering, and FontAwesome 6 icons.
- **Backend & API Layer:** Node.js as the non-blocking asynchronous event loop runtime; Express.js as the minimalist RESTful API framework; Express Router for modular controller mapping; Multer middleware for secure photo uploads; CORS and Body-Parser.
- **Database & Security:** MySQL 8.0 ACID-compliant relational database engine; mysql2 driver supporting connection pooling and promises; JSON Web Tokens (JWT) for stateless authentication; bcrypt.js for 10-round password hashing; Dotenv for secret management.
- **External APIs & Tools:** Razorpay Payment Gateway API for secure card/UPI transactions; Postman for REST endpoint testing; Visual Studio Code as the primary IDE; Git/GitHub for version control; and Nodemon for server hot-reloading.

Future Enhancements & Project Conclusion

Future Enhancements & Scalability:

- AI Computer Vision: Deploy deep learning models (YOLOv8 / ResNet) to scan returned items via camera and automate condition grading.
- Cross-Platform Mobile App: Build Flutter / React Native apps with automated push notifications for midnight price drops.
- In-Store QR Code Scanners: Place QR tags on physical bins so in-store walk-in customers can scan items on their phones.
- Real-Time Pallet Auctions: Integrate WebSockets via Socket.io to support real-time wholesale pallet auctions.

Project Conclusion & Takeaways:

✓ Successfully digitized an offline liquidation bin retail model into an automated 24/7 web e-commerce platform.

✓ Demonstrated a full-stack architecture with a responsive frontend, RESTful Node.js backend, and a 13-table 3NF MySQL database.

✓ Implemented real business utility: dynamic daily price drops, physical bin mapping, online payments, and gamification.

Thank You! — Questions & Answers Session Open for Discussion